Chapter III: Economic Growth

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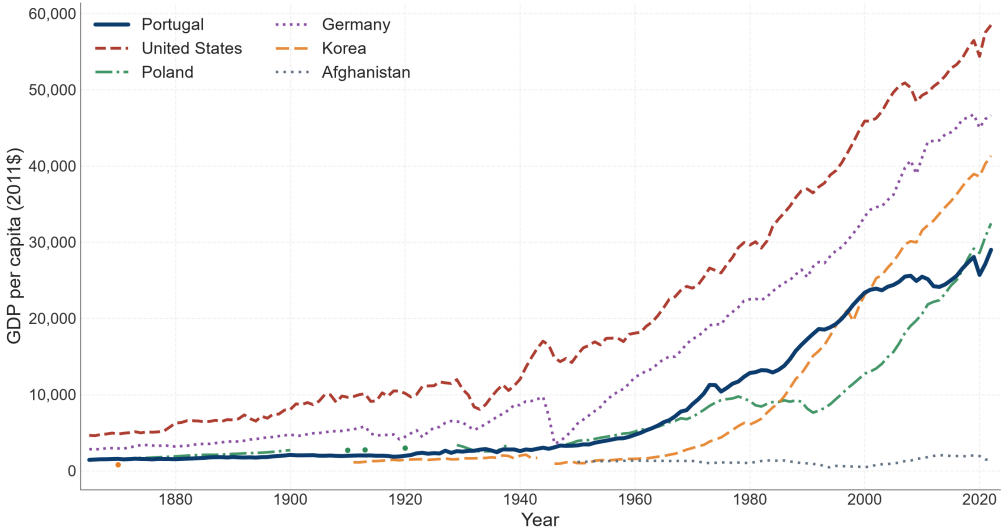
Why Study Economic Growth?

- As we saw in chapter I, GDP shapes to a large extent living standards.
- Even **small differences** in gdp growth rates create **huge income gaps** over time.
- Rule of thumb: if gdp grows at $g\%$ per year,

$$\text{Doubling time} \approx \frac{70}{g} \text{ years.}$$

- Examples:
  - **South Korea vs. Afghanistan**: similar income per capita in the 1960s, but sustained faster growth made Korea one of the richest countries in Asia while Afghanistan became x10 poorer.
  - **Poland vs. Portugal**: Portugal was nearly x3 richer in GDP per capita in the early 1990s, yet Poland's consistently higher growth has allowed it to surpass Portugal.
- Understanding what drives growth helps us understand why some countries prosper while others fall behind.

GDP per Capita Since 1865



Source: Maddison Project Database 2023 (Bolt and van Zanden et al.). Missing years left blank.

The Solow–Swan Model

- The Solow–Swan (1956) model is a simple model to study long-run economic growth.
- It provides a description of how capital accumulation and technology shape GDP per capita.
- In contrast to the model that we saw in chapter 2, the capital stock is going to be determined within the model.
- Let's see the main ingredients.

Production Function

- A production function tells us how inputs become output:

$$Y = F(K, L)$$

- Inputs:
  - K: capital (machines, buildings, equipment)
  - L: labor (hours of work)

Consumption and Investment

- The economy is closed and there is no government and population is constant: $\bar{L}$
- The national income identity simplifies to:

$$Y = C + I$$

- All output is either consumed or invested.

Savings and Investment

- The savings rate is exogenous and constant:

$$\frac{S}{Y} = s.$$

- Income is equal to output: Y is both.
- Savings: $S = Y - C$.
- Savings rate: $\frac{Y-C}{Y} = s$.
- In a closed economy, savings equal investment: $S = I$.

Depreciation and Capital Accumulation

- In contrast to the previous chapter, this model is going to be dynamic.
- The capital stock depreciates at a constant rate δ.
- Capital wears out: machines break, buildings deteriorate, technology becomes obsolete.
- Each period, a fraction δ of the capital stock loses its productive value.

$$K_{t+1} = (1 - \delta)K_t + I_t$$

- $(1 - \delta)K_t$: capital that survives depreciation.
- I_t: new capital created through investment.

Capital Accumulation: A Numerical Example

- Consider an economy with initial capital $K_0 = 1,000$ units, constant investment $I_t = 200$ units, and depreciation rate $\delta = 0.10$.
- How does the capital stock evolve?

Time t	Capital K_t	Investment I_t	Depreciation δK_t	Change ΔK_{t+1}
0	1,000	200	100	100
1	1,100	200	110	90
2	1,190	200	119	81
3	1,271	200	127	73
4	1,344	200	134	66
5	1,410	200	141	59

- Each period: $\Delta K_{t+1} = I_t - \delta K_t$.
- Net investment decreases over time as depreciation rises with the capital stock.

Per Capita Production Function

We rewrite the production function in per-capita terms. Define:

$$y \equiv \frac{Y}{\bar{L}}, \qquad k \equiv \frac{K}{\bar{L}}$$

where y is GDP per capita and k is capital per worker.

Starting from the aggregate production function:

$$y = \frac{F(K, \bar{L})}{\bar{L}} = F\left(\frac{K}{\bar{L}}, 1\right) \equiv f(k)$$

- The first step substitutes $Y = F(K, \bar{L})$ from the aggregate production function.
- Because of constant returns to scale, scaling both inputs by $1/\bar{L}$ leaves output per worker well-defined.
- We define the per-capita production function $f(k)$ as the output produced by one worker with k units of capital.

Dynamics of Capital per Worker

We want to understand how capital per worker evolves over time:

$$k_{t+1} \equiv \frac{K_{t+1}}{\bar{L}}$$

Start from the capital accumulation equation:

$$K_{t+1} = (1-\delta)K_t + I_t$$

Use:

$$I_t = sY_t$$

Combining these:

$$k_{t+1} = (1-\delta)k_t + sy_t$$

$$\Rightarrow \Delta k_{t+1} = (1-\delta)k_t + sy_t - k_t = sf(k_t) - \delta k_t$$

Interpretation of the Dynamics

The evolution of capital per worker depends on two opposing forces:

- **Investment per worker:**

$$\underbrace{sf(k)}_{\text{adds to } k}$$

- **Depreciation:**

$$\underbrace{\delta k}_{\text{subtracts from } k}$$

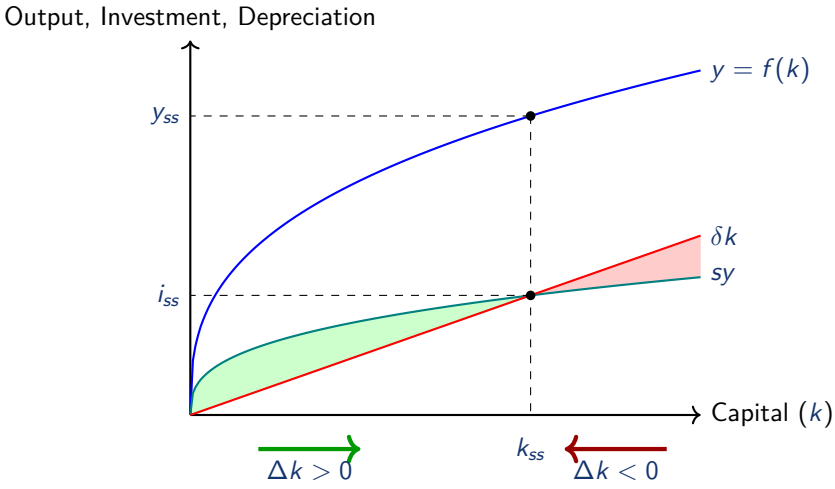
  Depreciation consumes part of the capital stock.

Thus:

$$\Delta k > 0 \iff sf(k) > \delta k$$

$$\Delta k < 0 \iff sf(k) < \delta k$$

Solving the model



The Steady State

The steady state is where capital per worker does not change:

$$k_{t+1} = k_t = k_{ss}$$

Condition:

$$sf(k_{ss}) = \delta k_{ss}$$

- If $k < k_{ss}$: investment exceeds depreciation+dilution $\Rightarrow$ k rises.
- If $k > k_{ss}$: investment is too small to sustain the level of capital $\Rightarrow$ k falls.
- The economy converges to k_{ss} over time.

Closed-Form Steady State: Cobb–Douglas

If:

$$y = Ak^{\alpha}$$

where A denotes a productivity parameter driving higher or lower output given a quantity of capital.

Then, steady state satisfies:

$$sAk_{ss}^{\alpha} = \delta k_{ss}$$

Solving:

$$k_{ss} = \left(\frac{sA}{\delta}\right)^{\frac{1}{1-\alpha}}$$

$$y_{ss} = A\left(\frac{sA}{\delta}\right)^{\frac{\alpha}{1-\alpha}}$$

Why are poor countries poor?

- Now let's use our model to try to rationalize differences in GDP per capita across countries and what could be its main drivers.
- Let's imagine that we have two economies:
  - Poor economy: $s_{\text{poor}}, A_{\text{poor}}$
  - Rich economy: $s_{\text{rich}}, A_{\text{rich}}$
- Assume that the depreciation rate is equal across these two countries.
- The ratio of GDP per capita across these two economies is given by:

$$\frac{y_{\text{rich}}}{y_{\text{poor}}} = \left(\frac{A_{\text{rich}}}{A_{\text{poor}}}\right)^{\frac{1}{1-\alpha}} \left(\frac{s_{\text{rich}}}{s_{\text{poor}}}\right)^{\frac{\alpha}{1-\alpha}}$$

Why are poor countries poor?

The Role of Investment vs Productivity differences

- With $\alpha = 1/3$, we get:

$$\frac{y_{\text{rich}}}{y_{\text{poor}}} = \left(\frac{A_{\text{rich}}}{A_{\text{poor}}}\right)^{3/2} \left(\frac{s_{\text{rich}}}{s_{\text{poor}}}\right)^{1/2}$$

- The richest countries in the world have a GDP per capita around 70 times larger than poor countries
- Investment rates are around 28% in rich countries and 7% in poor countries
- These means that according to the Solow model, differences in savings rate would make rich countries around $\left(28/7\right)^{1/2} = 2$ times richer
- If poor countries had the productivity of rich countries, they would be 35 times richer! (in contrast to 14 in the previous chapter, why?)

Economic Growth in the Solow Model

Steady State

- The Solow model implies convergence to a **steady state**.
- In the long run:
  - Capital and output are constant: K^*, Y^*.
  - Output per person is constant:

$$y^* \equiv \frac{Y^*}{\bar{L}}$$

  - Consumption per person is constant:

$$c^* = (1 - s)y^*$$

- Long-run growth in y is zero.

Economic Growth in the Solow Model

Transition vs. Data

- The model allows for growth only **during transition**.
- If $K_0 < K^*$:
  - Capital accumulates.
  - Output per person grows temporarily.
- Once $K = K^*$, growth stops.
- But in the data:
  - GDP per capita grows persistently over time.
  - U.S. GDP per capita: $\approx 2\%$ per year for over a century.

Economic Growth in the Solow Model

Key Lesson

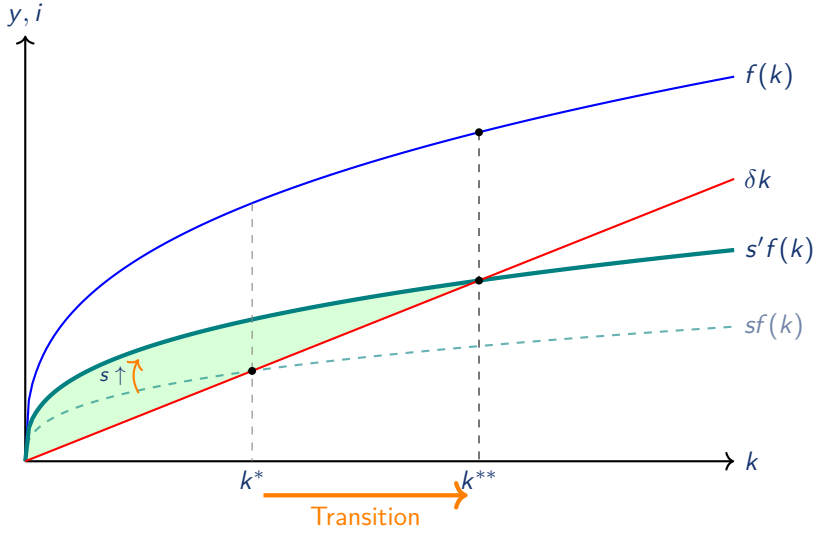
- Capital accumulation cannot explain **long-run** growth.
- Diminishing returns to capital imply:
  - Falling marginal product of capital.
  - Investment eventually only offsets depreciation.
- Physical capital drives **medium-run** growth, not long-run growth.
- Something else must sustain growth.

Increase in the Savings Rate

Suppose an economy in steady state k^* experiences a permanent increase in the savings rate from s to s'.

- The investment curve $sf(k)$ rotates upward to $s'f(k)$. For any given level of capital, the economy now saves and invests more.
- At the old steady state k^*, investment now exceeds depreciation: $s'f(k^*) > \delta k^*$.
- This surplus causes the capital stock to grow ($\Delta k > 0$).
- As k rises, output y also rises along the production function.
- Growth continues but slows down as the economy approaches k^{**}, where $s'f(k^{**}) = \delta k^{**}$.
- The levels of capital per worker and output per worker are permanently higher, but the long-run *growth rate* eventually returns to zero.

Increase in Savings Rate

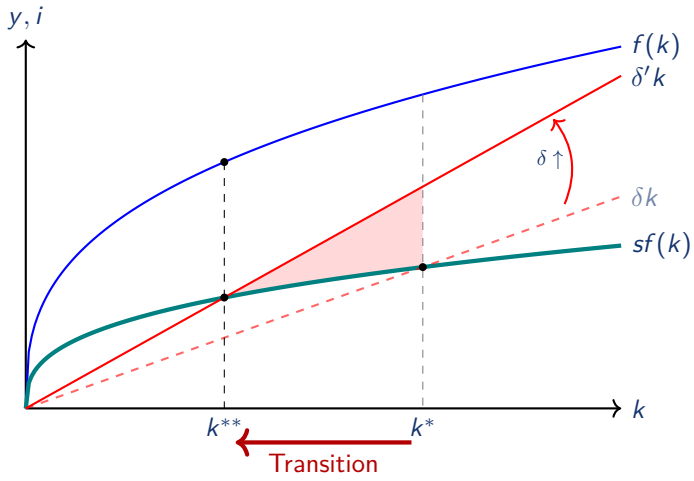


Increase in the Depreciation Rate

Suppose an economy in steady state k^* faces a permanent increase in the depreciation rate from δ to δ'.

- The investment line δk rotates upward (becomes steeper). For every unit of capital, more investment is now required just to maintain the current level.
- At the original steady state k^*, depreciation now exceeds investment: $\delta' k^* > sf(k^*)$.
- The net change in capital becomes negative ($\Delta k < 0$).
- The capital stock begins to decline, which in turn reduces output per worker y along the production function.
- The capital stock continues to contract until it reaches k^{**}, where the lower level of investment is once again sufficient to cover the higher depreciation rate.
- Both capital per worker and output per worker settle at permanently lower levels ($y^{**} < y^*$).

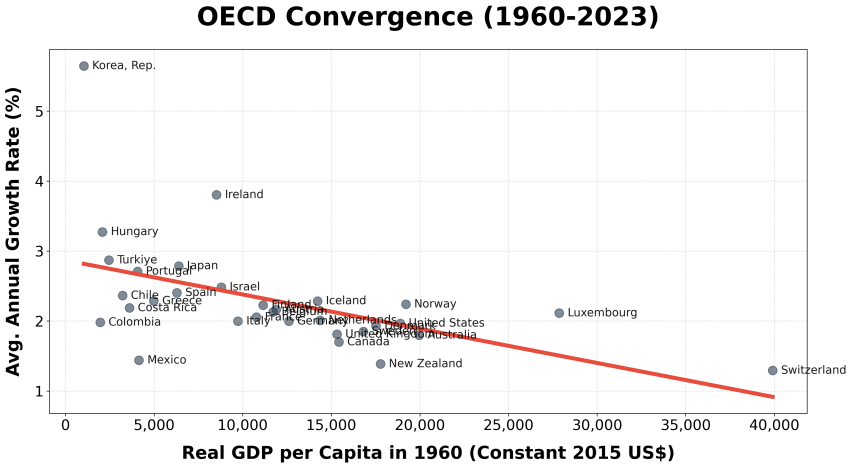
Comparative Statics: Increase in Depreciation



Transitional Dynamics

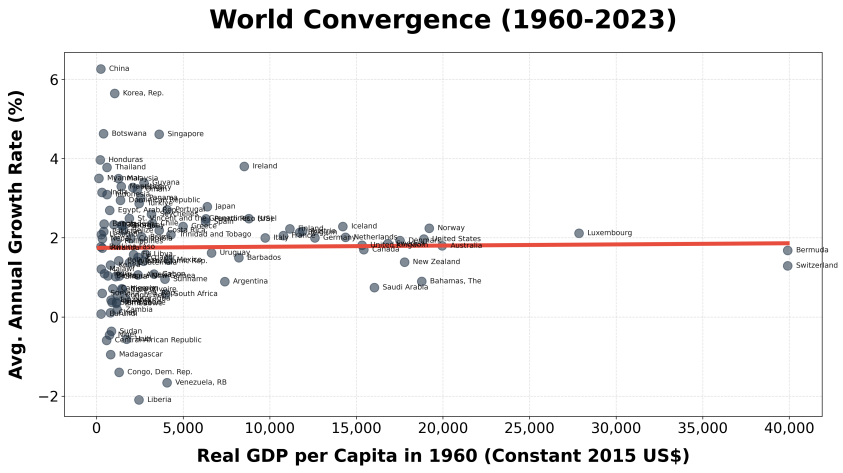
- The Solow model provides a framework for understanding why some countries grow faster than others based on their position relative to their steady state.
- The farther below its steady state an economy is, the faster it will grow.
- As k approaches k_{ss}, the growth rate of output per capita gradually declines toward zero.
- The Solow model predicts **convergence:**
  - Countries with similar steady states should eventually converge
  - Poorer countries should grow faster.
  - China can be viewed as an economy far below its steady state catching up.

Theory vs Data



Source: World Bank WDI. Growth = avg. annual log change.

Theory vs Data



Source: World Bank WDI. Growth = avg. annual log change.

Case Study: Korea vs Afghanistan

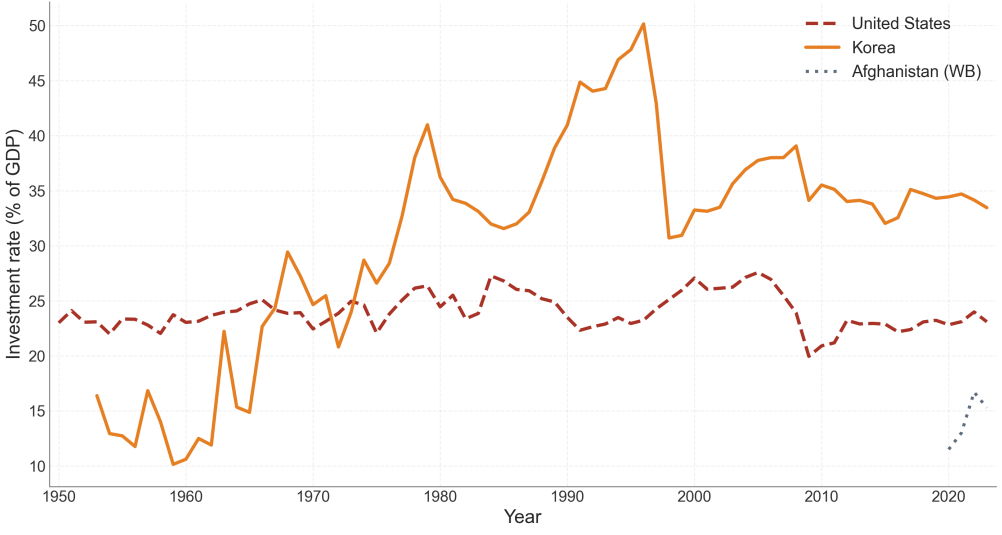
- We started the chapter by asking: Why has South Korea grown so much faster than Afghanistan in the past half century?
- We can apply what we have learned on transitional dynamics to try to rationalize this difference in growth performance.
- We can think that changes in investment and productivity have pushed South Korea to a much higher steady state than Afghanistan.
  - Starting from around 10 percent of U.S. income in 1960, maybe Korea's steady-state level moved up to 50 or 75 percent while the Afghanistan's level remained at 10 percent.
- This would entail that in 1960, Korea was far below its steady state and thus had a lot of room to grow
- Afghanistan was already close to its steady state and thus had little room to grow.

Case Study: Korea vs Afghanistan

- What does the Solow model tell us about the determinants of the steady- state ratio of income in South Korea to income in the United States?
- Assuming these economies have the same rate of depreciation, the ratio of their steady-state incomes is given by:

$$\frac{y^{Korea}}{y^{US}} = \left(\frac{A^{Korea}}{A^{US}}\right)^{3/2} \left(\frac{s^{Korea}}{s^{US}}\right)^{1/2}$$

Investment Rate: US, Korea, and Afghanistan



Source: Penn World Table 11.0 (pwt110.dta). Variable: csh_i (% share of investment in GDP).

Case Study: Korea vs Afghanistan

- The U.S. investment rate is relatively stable
- The investment rate in Korea rises dramatically, from about 15 percent in the 1950s to around 40 percent in the early 1990s.
- According to the Solow model, this rise should significantly increase Korea's steady- state income
- Transitional dynamics tells us that the Korean economy will grow rapidly as it makes the transition to its new, higher level.
- On the other hand, the investment rate in Afghanistan remains at much lower levels for the data that I could find.
- We could think that the steady is trapped at a low level and thus the economy has little room to grow but hopefully the recent increases in investment rate will help the country to grow faster in the future.

What Does the Solow Model Teach Us?

- The Solow framework is a key building block of macroeconomics of growth:
  - A production function with capital and labor.
  - An accumulation equation linking saving today to capital tomorrow.

- **Strengths**:
  - Explains income levels in the long run (investment, TFP, depreciation).
  - Explains growth differences through transition dynamics.

- **Limitations**:
  - Investment differences explain only a small part of income gaps.
  - TFP differences remain largely unexplained.
  - No theory of sustained long-run growth.

- Capital accumulation drives medium-run growth, but long-run growth must come from elsewhere.

Beyond the Solow Model

- The Solow model does not provide a theory of long-run growth.
- Once the economy reaches its steady state, growth stops because of decreasing returns to capital.
- Paul Romer (1990) argued that *ideas* are fundamentally different from capital and can generate non-decreasing returns.
- Understanding this difference between ideas and capital/labor lets us build a model with long-run growth.

Rival vs. Non-Rival Goods

- **Rival goods**: use by one agent *excludes* use by another.
  - Examples: machines, workers, raw materials.
  - If a firm uses a machine, no other firm can use the same machine simultaneously.
- **Non-rival goods**: use by one agent does *not* prevent use by another.
  - Examples: the recipe for Bolo de Bolacha from your grandmother, blueprints, software, scientific formulas.
  - Newton's laws can be applied by every scientist simultaneously.
- **Ideas are non-rival**: once discovered, they can be used by everyone.

The Romer Model: Setup

- Paul Romer (1990) developed a model of long-run growth based on the idea that ideas are non-rival goods.
- In this model A_t is going to represent the stock of ideas at time t.
- In contrast to the Solow model, A_t is not exogenously given and fixed, but rather it is endogenously determined by the economy and grows over time.
- The Romer model has two sectors: a final goods sector and a research and development (R&D) sector.
- The final goods sector produces output using labor and ideas.

The Romer Model: Setup

- Final goods sector:

$$Y_t = A_t L_Y$$

  Output depends on the stock of ideas A_t and labor allocated to production L_Y.

- R&D sector:

$$A_{t+1} - A_t = \delta_A A_t L_A$$

  The production of new ideas depends on the existing stock of ideas A_t and labor allocated to research L_A, where δ_A is a productivity parameter.

- We assume that a constant fraction ℓ of the labor force ($\bar{L}$) is allocated to R&D:
  - $L_Y = (1 - \ell)\bar{L}$
  - $L_A = \ell \bar{L}$
  - Total labor constraint: $L_Y + L_A = \bar{L}$

Solution of the Romer Model

- Output per person:

$$y_t = \frac{Y_t}{\bar{L}} = A_t(1 - \ell)$$

- Growth rate of ideas:

$$g_A = \frac{A_{t+1} - A_t}{A_t} = \delta_A L_A = \delta_A \ell \bar{L}$$

  The growth rate of ideas is going to depend on

  - the number of researchers: $L_A = \ell \bar{L}$
  - the productivity of research: δ_A

- Ouput per person also grows that the same constant rate.

$\Rightarrow$ **A Theory of Sustained Growth**

Why there is sustained growth in the Romer model?

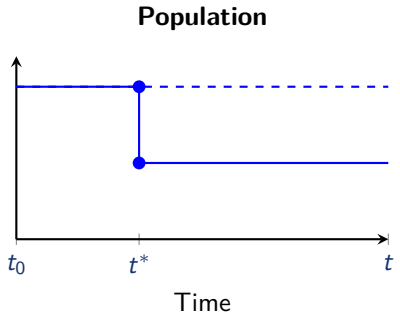
- In Solow, the accumulation of capital runs into diminishing returns: each additional unit of capital adds less and less to output.
- Eventually, investment only offsets depreciation and growth stops.
- In Romer, the accumulation of ideas does not run into diminishing returns:

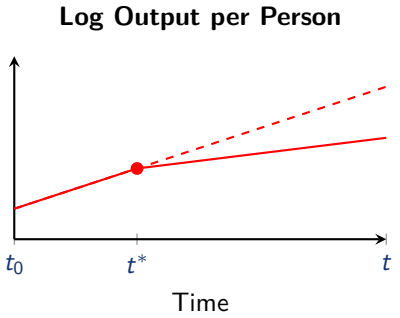
$$A_{t+1} - A_t = \delta_A A_t L_A$$

- As we accumulate more knowledge, the return to knowledge does not fall.
- Old ideas continue to help us produce new ideas in a virtuous circle that sustains economic growth.

Lower Population

- A decrease in population $\bar{L} \to \tilde{L}$ decreases the number of researchers $L_A = \ell \bar{L}$, which slows down the growth rate of ideas $\tilde{g}_A = \delta_A \ell \tilde{L}$.
- This is the **scale effect**: larger economies grow faster because they have more researchers.

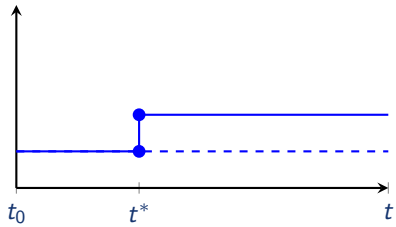




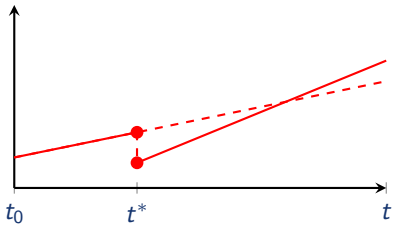
Higher Fraction of Researchers

- An increase in the fraction of researchers $\ell \to \tilde{\ell}$ increases the growth rate of ideas $\tilde{g}_A = \delta_A \tilde{\ell} \bar{L}$ and output.
- But as people are allocated to research, less people are allocated to production, so the level of output per person falls.
- Increasing the research share involves a trade-off.

Fraction of Researchers



Log Output per Person



Conclusion

- The Solow model explains medium-run growth through capital accumulation, but long-run growth must come from elsewhere.
- The Romer model provides a theory of sustained long-run growth.
- Because ideas are nonrivalrous, it is not ideas per person that matter, but rather this total stock of ideas.
- Increases in the stock of knowledge lead to sustained economic growth for countries that have access to that knowledge.